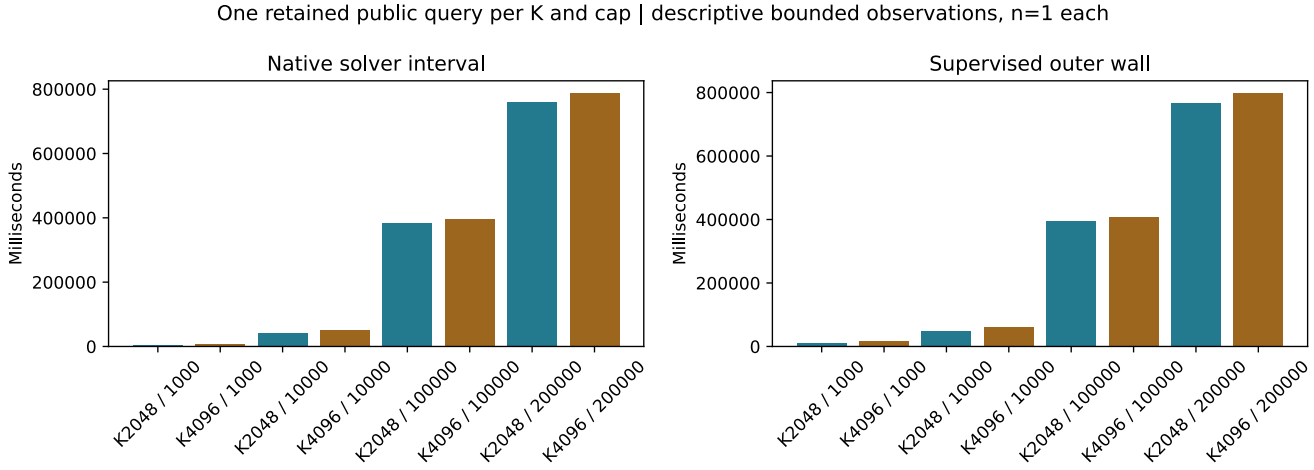
Native N83 factor-base continuation: 200000-conflict caps

Public synthetic curve icv1-f2m83-tm6151469093347-debefd74, a=0 over x^83+x^45+x^2+x+1; subgroup order24178516392307962166856 89. Negative-trace primary curve, distinct from the positive-trace N83 cold holdout. Full protocol, source identity, run records and S3 verification links are in REPORT.md beside this PDF.

The user requested another higher conflict cap for both bases and complete raw upload. The separately committed protocol raises each cap100000→200000, with1200seconds per worker and1290seconds per mechanical command. K2048 runs once, then K4096 after assessment. Maximum new worker allocation2400seconds; readiness, build/test, replay, rendering and storage are separate overhead. Earlier1000/10000/100000-conflict observations remain immutable and are not pooled as repetitions.



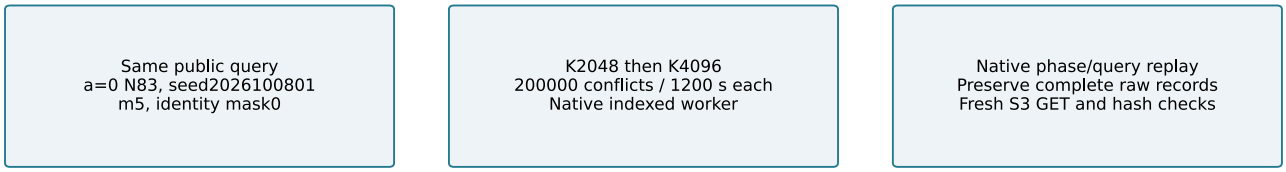
Native solver intervals are derived in Rust as solver_return.elapsed_ms minus solver_enter.elapsed_ms. Outer wall includes supervision/setup and is a different interval. A censored prefix supplies no terminal counter or complete solver duration. These are descriptive n=1 bounded observations per K/cap, without an A/A interval, timing-rank admission, external fastest-tool claim or IC/rho ratio.

Recorded new outcomes (native audit; n=1 each): K2048: PASS_NATIVE_BOUNDED_ZERO_RELATION_REPLAY; terminal conflicts 200000, models 0, relations 0, rank 0; solver 759947.84721 ms, outer 765409.242176 ms. K4096: PASS_NATIVE_BOUNDED_ZERO_RELATION_REPLAY; terminal conflicts 200000, models 0, relations 0, rank 0; solver 786700.6455450001 ms, outer 797133.398274 ms.

Recorded outcomes, guard validation and custody

K / cap	Conflicts	Native audit	Solver ms
K2048 / 1000	1000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	4861.868809
K4096 / 1000	1000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	7814.674415
K2048 / 10000	10000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	42413.639201
K4096 / 10000	10000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	50433.062636
K2048 / 100000	100000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	383937.390858
K4096 / 100000	100000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	395987.838978
K2048 / 200000	200000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	759947.847210
K4096 / 200000	200000	PASS NATIVE BOUNDED ZERO RELATION REPLAY	786700.645545

Frozen higher-cap continuation | prior trials are separate observations



Fresh v8 cards reserve core14/30; v7 preflight refusal is preserved in this round.

Retained natural worker SHA2563a31bd2bedee5519d30b517f39ec27f72561ce898213657688c55b51bbcdf7b6 is not rebuilt. All9watched solver/encoding/example/adapter sources match source660f0f4b10. New source7f07562459 pins guard/run metadata; it is not a claim that the retained worker was built at that HEAD. The original full worker/source archive remains content-addressed in the upstream100K/XOR publications. Same single public query, seed2026100801, m5 and identity mask0.

New immutable guard-v7 SHA25621bb3a1e824d5e2e8de4b94960d1ebe1b0676b4f5904b7cb58095c6e0594e6e1 only extends internal natural admission to200000conflicts/1200s. Native witness100000/900 and external1000000/900 limits remain unchanged. New budget-boundary tests and existing native harness checks pass:16unit+7card tests. Earlier library4398-pass/118-ignored validation and post-base-merge38module/19exporter/2model checks are inherited, with the original socket-denial/rerun qualifications preserved; no new all-library rerun is claimed. Guard-v8 SHA2562ec47f22528893777941de5dc4a68e8b2ac02d920c87449ed835137ddb2c0273 adds an optional strict full-core CPU list;17unit+7card tests pass. Fresh cards reserve14/30 with unchanged quiet and eviction gates. The v7 attempt refused CPU15/31 before launch; the first v8 Docker create RPC timed out before launch, and a delayed Created/Pid0 container was inspected and removed. Both failures and the later cleanup correction remain raw evidence; no timed solve was retried.

Same immutable Docker image541096ae9b05…,1CPU quota/4GiB/no extra swap/networknone/readonly root-source-input-worker/capdropALL/no-new-privileges/64PIDs. Exclusive native lock, CPU/sibling reservation, movable-user eviction/restoration, unchanged quiet thresholds and bounded pre-launch readiness wait. Actual resource/CPU/quiet/cleanup records are retained. Fixed kernel threads and missing worker context-switch/fault accounting prevent automatic L2 admission. No builds/tests/heavy work from this task runs during measured arms.

Director compiles/assesses; mechanical executes exact frozen prompts with actual stdout captured under noclobber and fresh paths. Native replay checks original-field/group queryaG+bQ, phase order, source/input/ELF/stream custody, caps, exit/OOM and cleanup. Positive relations remain fail-closed until original-group/rank/scalar replay supports them. Unknown is not UNSAT or absent decompositions; zero yield at a finite cap cannot establish the best base.

Only native tools run research/replay/derive results. Python lays out recorded values and inspects PDFs. Same-source SVG/150dpiPNG and benchmark copies are included. This is a bounded diagnostic with no solver algorithm change or admitted speed claim. REPORT.md lists canonical cold/rho/identity graphs left unchanged and complete raw S3 archive/fresh-download checks.

Analysis-v9 corrects the draft Docker-launch label;18unit+7card tests pass. Measured runs are unchanged. Both setup failures remain archived and described separately.